

01 Obtaining and Configuring the Large Model API Key

1. OpenAI Account Registration and Setup

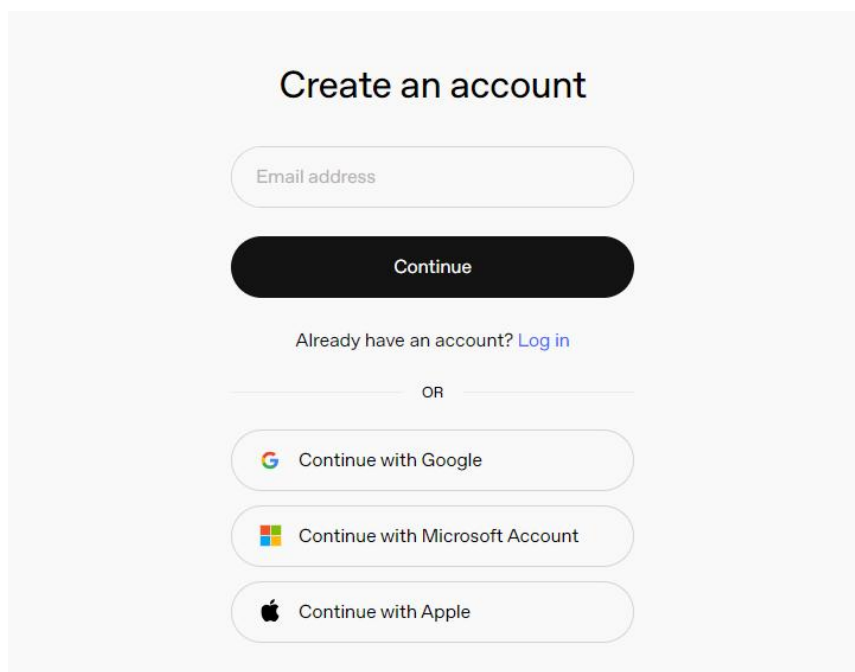
1.1 Registration

- 1) Visit the following link to access the OpenAI platform:

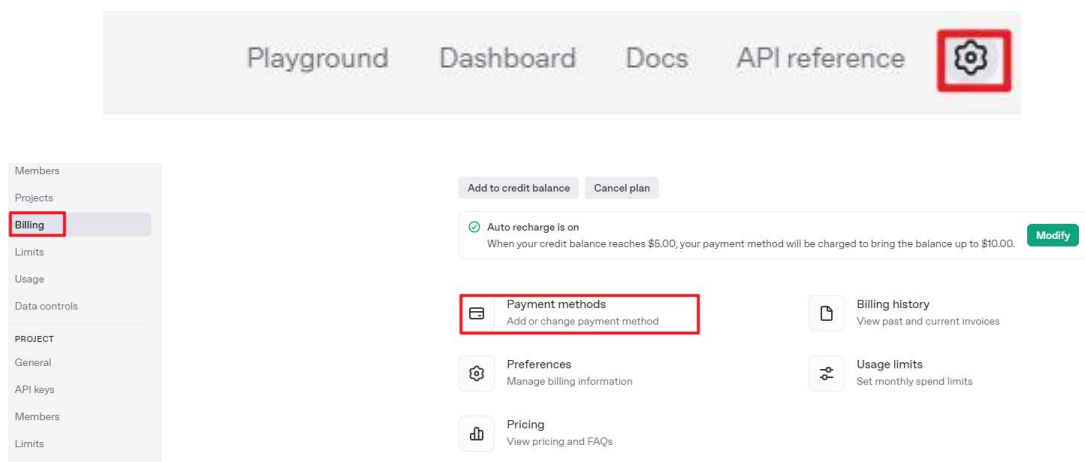
<https://platform.openai.com/docs/overview>



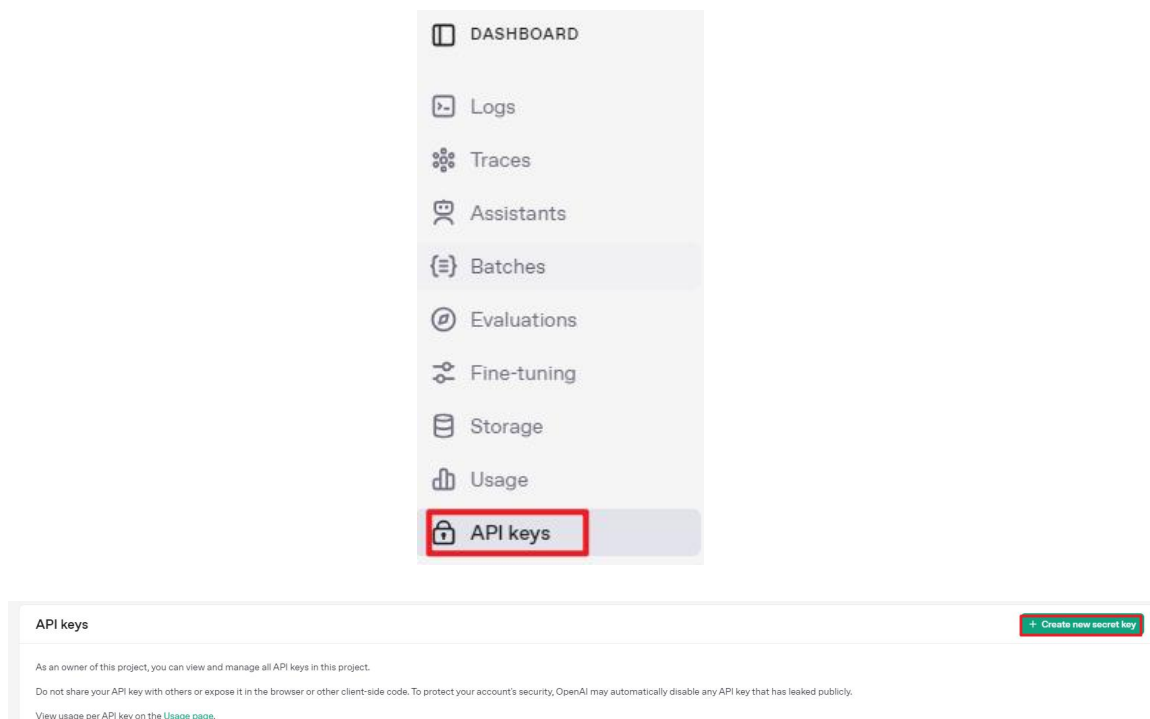
- 2) You can register using your Google, Microsoft, or Apple account as prompted.



- 3) Once registered, click on the Settings icon, navigate to Billing, and then select Payment Methods to add a payment option. This is required to purchase tokens.



- 4) After setting up your billing information, go to the API Keys section, click Create New Secret Key, and follow the on-screen instructions. Be sure to save the key for future use.



Create new secret key

Owned by

☐ You ☐ Service account

This API key is tied to your user and can make requests against the selected project. If you are removed from the organization or project, this key will be disabled.

Name Optional

Project

Permissions

☐ All ☐ Restricted ☐ Read only

With these steps completed, you've successfully created and deployed your large model API key, which will be used in the upcoming course sessions.

1.2 Deploying the API Key

The configuration file of this program is saved in:

/home/ubuntu/ros2_ws/src/large_models/large_models/large_models/config.py

- 1) Use the following command and press Enter to navigate to the directory containing the configuration file:

```
cd /home/ubuntu/ros2_ws/src/large_models/large_models
```

```
ROS_DOMAIN_ID: 90
VERSION: [V2.0.0|ROS2 2.0|2025-02-12]
> cd /home/ubuntu/ros2_ws/src/large_models/large_models
~ /ros2_ws/src/large_models/large_models | git main !499 ?49
```

- 2) Enter the following command to open the configuration file and insert the OpenAI key:

gedit config.py

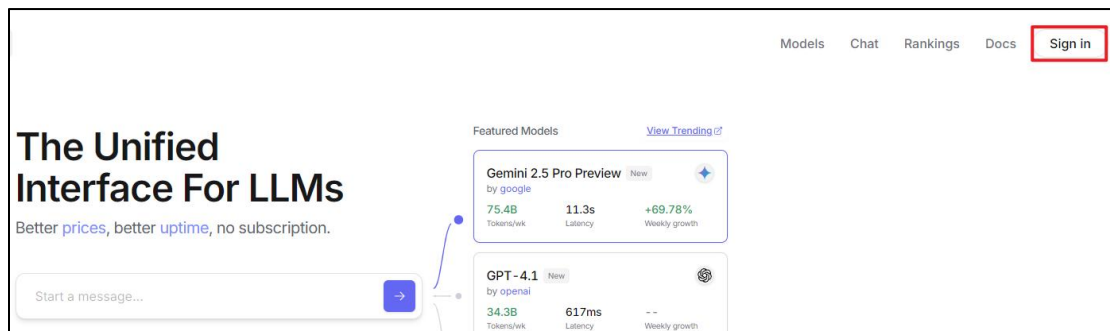
```
> cd /home/ubuntu/ros2_ws/src/large_models/large_models
└─ /ros2_ws/src/large_models/large_models | git main !499 ?50
└─ gedit config.py
```

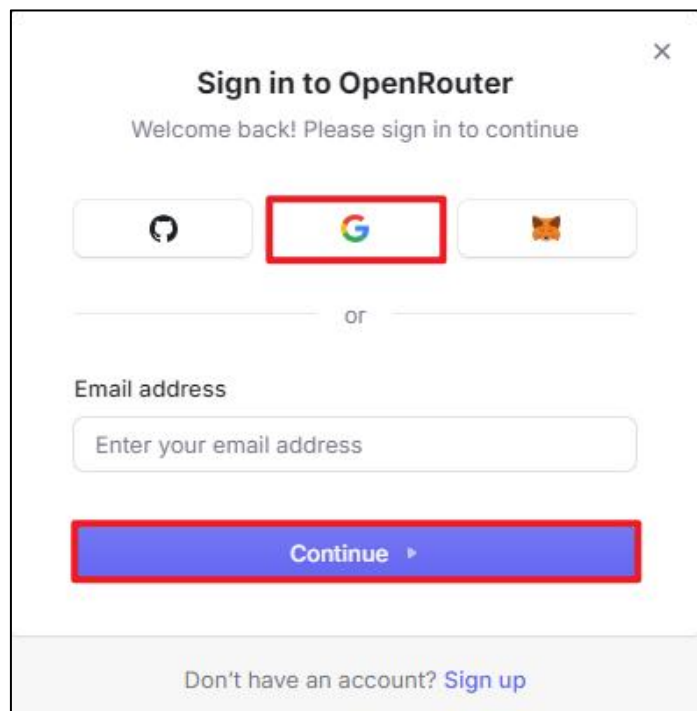
```
24 ###Internationally###
25 vllm_api_key = ''
26 vllm_base_url = 'https://openrouter.ai/api/v1'
27 vllm_model = 'qwen/qwen2.5-vl-72b-instruct:free'
28
29 llm_api_key = ''
30 llm_base_url = 'https://api.openai.com/v1'
31 llm_model = 'gpt-4o-mini'
32 openai_vllm_model = 'gpt-4o'
33 openai_tts_model = 'tts-1'
34 openai_asr_model = 'whisper-1'
35 openai_voice_model = 'onyx'
36 #####
37
```

2. OpenRouter API Configuration

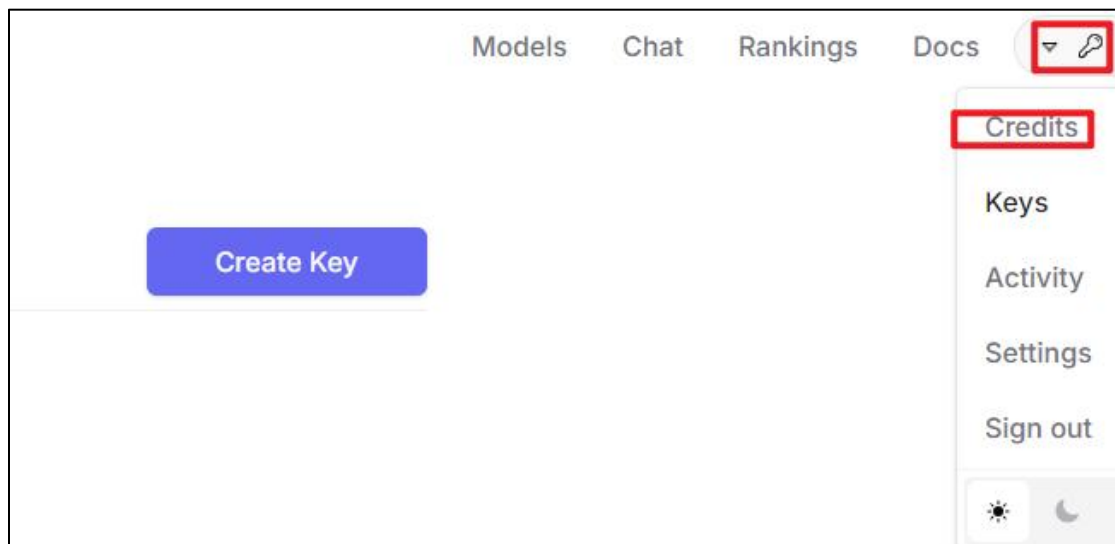
1) Visit the following link to access the OpenAI platform: <https://openrouter.ai/>

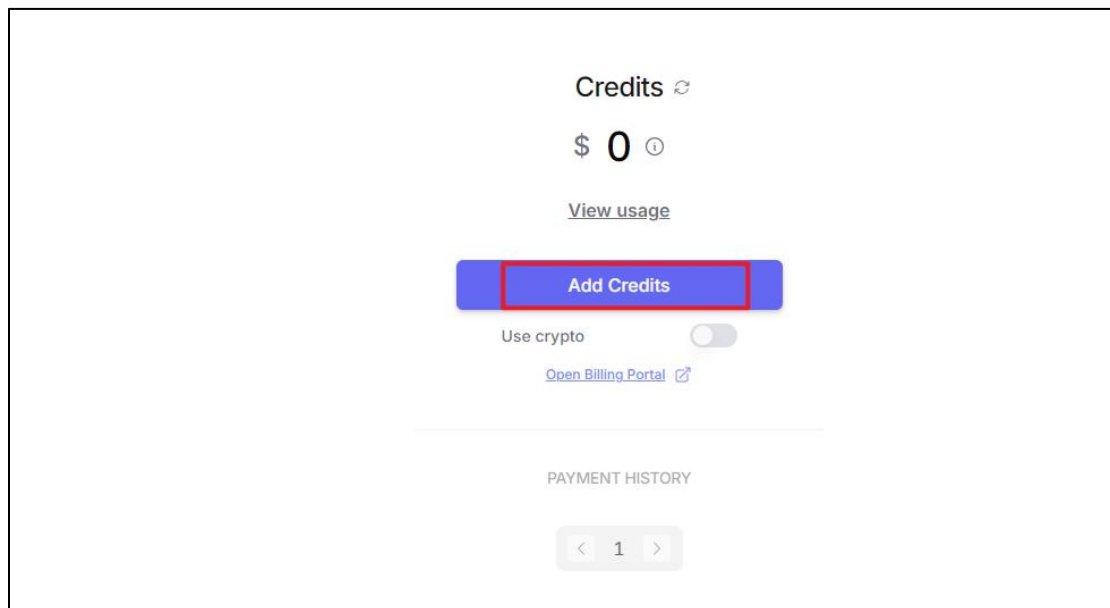
Click on "Login" and use your Google account or another supported account to register or log in.



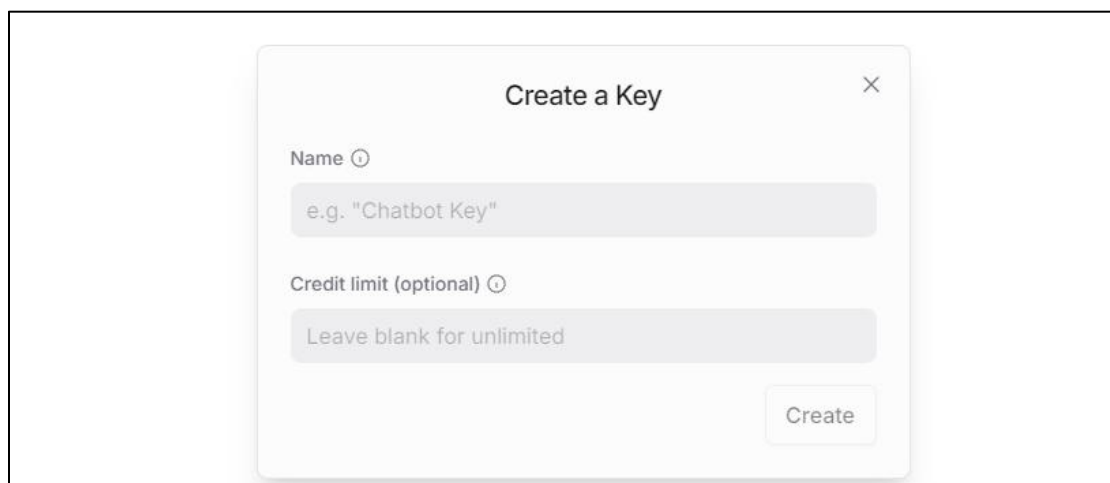
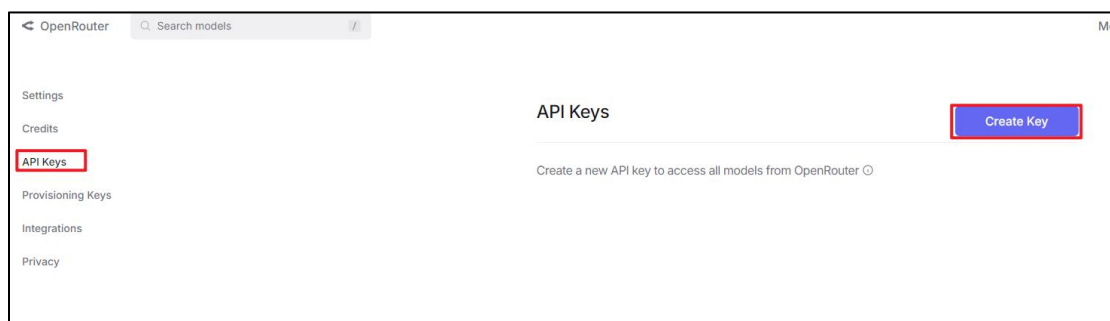


- 2) After logging in, click on the icon in the top-right corner, select "Credits," and link your payment method.





- 3) To create an API key: Navigate to "API Keys" and click "Create Key." Then follow the prompts to generate your API key, and make sure to save it for future use.



- 4) Use the following command and press Enter to navigate to the directory

containing the configuration file:

```
cd /home/ubuntu/ros2_ws/src/large_models/large_models
```

```
ROS_DOMAIN_ID: 90
VERSION: [V2.0.0|ROS2 2.0|2025-02-12]
> cd /home/ubuntu/ros2_ws/src/large_models/large_models
└─ ~/ros2_ws/src/large_models/large_models | git main !499 ?49 ▶
```

- 5) Open the configuration file using the following command, then paste your OpenRouter API key into it:

```
gedit config.py
```

```
> cd /home/ubuntu/ros2_ws/src/large_models/large_models
└─ ~/ros2_ws/src/large_models/large_models | git main !499 ?50 ▶
  gedit config.py
```

```
22 #####
23
24 ##Internationally##
25 vllm_api_key = ''
26 vllm_base_url = 'https://openrouter.ai/api/v1'
27 vllm_model = 'qwen/qwen2.5-vl-72b-instruct:free'
28
29 llm_api_key = ''
30 llm_base_url = 'https://api.openai.com/v1'
31 llm_model = 'gpt-4o-mini'
32 openai_vllm_model = 'gpt-4o'
33 openai_tts_model = 'tts-1'
34 openai_asr_model = 'whisper-1'
35 openai_voice_model = 'onyx'
36 #####
37
```